

Ultrasonic Distance Collection

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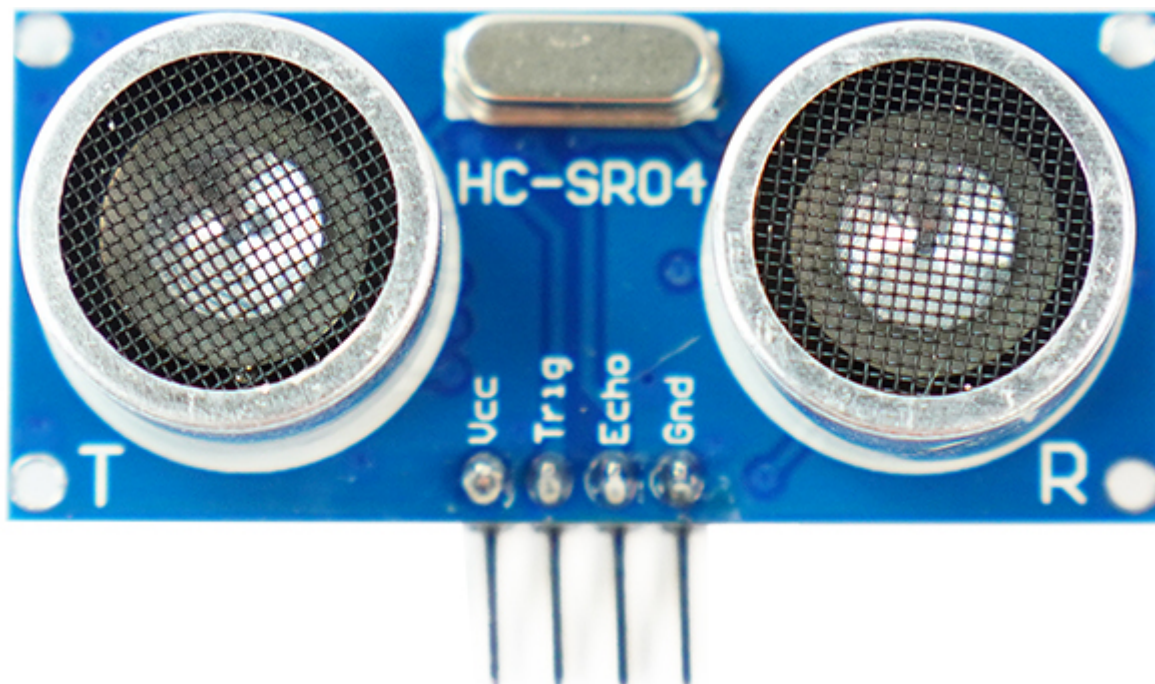
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Experimental Purpose

The front of the Lastbit car is equipped with an ultrasonic sensor. In this section, we will learn how to read the data from the ultrasonic sensor.

The ultrasonic sensor is a sensor developed based on the characteristics of ultrasonic waves.

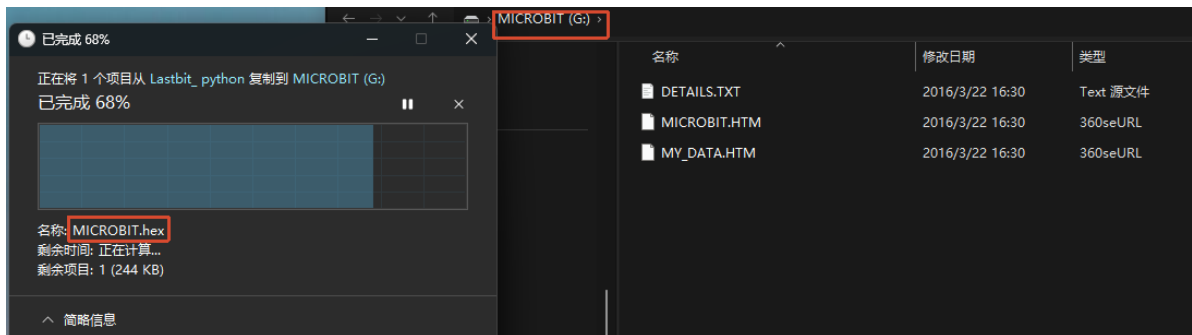
During operation, the `TRIG` pin emits an ultrasonic signal. After the ultrasonic signal is blocked by an obstacle, it returns. When the receiving pin `ECHO` receives the ultrasonic signal, it transmits the signal to the controller for processing. The controller calculates the time taken from transmitting to receiving the signal, thereby determining the distance between the probe and the object. The frequency range of sound waves that the human ear can hear is 20 Hz to 20 KHz. The frequency of the sound emitted by the ultrasonic module is greater than 20 KHz, so the human ear cannot hear the sound of the ultrasonic module.





Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

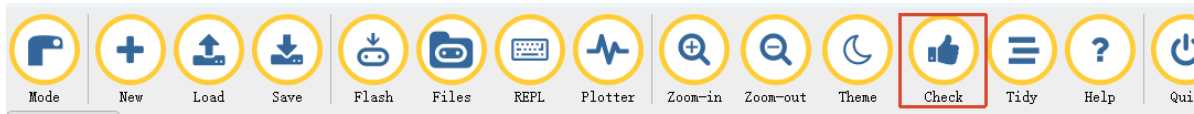
Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file.
You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

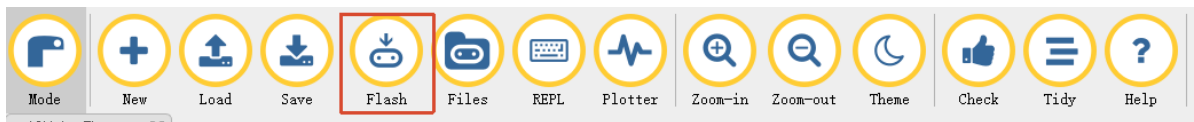
4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 1.Ultrasonic_Distance_Collection

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()
# Initialize servo angle
Lastbit.set_all_servos(90)
sleep(100)
# Lift up the robotic arm
# To avoid blocking the ultrasonic sensor
Lastbit.set_servo('S2', 140)
sleep(100)

while True:
    # Collect ultrasonic distance
    distance_mm = Lastbit.ultrasonic_distance()
    # If you need centimeters
    # distance_cm = round(distance_mm / 10)
    # Display ultrasonic distance
    display.scroll(distance_mm)
```

`ultrasonic_distance()` : reads the distance measured by the ultrasonic sensor. The return value is the distance information, in mm. If you need centimeters, you can convert the unit.

`display.scroll()` : scrolls to display the data.

`round()` : rounds the result to an integer.

Experimental Phenomenon

The Microbit LED matrix will continuously scroll to display the distance measured by the ultrasonic sensor.

Precautions

Make sure the car has sufficient battery. If -1 is returned, make sure there is an obstacle not far in front of the ultrasonic sensor, the wiring is not loose, and the power switch is in the ON position (switched to ON).